

Take-Home Case Study

Redcare Pharmacy

Time Advised 3-4 hours | Format Deck or doc + working prototype | Submission over email

Context

Every successful e-commerce experience depends on delivery. In pharmacy, delivery is not only a logistics step. It is part of the trust experience.

Customers want to know when their order will arrive before they commit to buying. A vague delivery estimate can reduce confidence, especially when customers are buying health-related products they may need soon. At the same time, an overly aggressive promise can damage trust if the order arrives late or if the promise changes during checkout.

This case has two parts. State your assumptions clearly, prioritise depth over breadth, and treat each scenario as if you have just joined the team.

1 Scenario: Delivery Promise Accuracy vs. Conversion

SCENARIO

Redcare currently shows broad delivery estimates such as "1-3 working days" across many customer touchpoints. This is operationally safe, but not very differentiated and may reduce customer confidence during product discovery and checkout.

The Business team wants to show more precise delivery promises earlier in the journey, especially on PDP, basket, and checkout. The hypothesis is that customers are more likely to convert when they know when the order will arrive.

The Data team has developed a new delivery prediction API. The first version works only for orders fulfilled from the Sevenum warehouse, which covers the majority of Redcare order volume. For around 30% of Sevenum-served products, the model predicts a shorter delivery time than the current generic 1-3 working day estimate.

However, the model also identifies smaller cohorts where the current estimate may be too optimistic. For some products, postcodes, carriers, or basket combinations, the model predicts a more realistic window such as "2-3 working days" or "3-5 working days".

Last Mile wants the API to be used symmetrically: if Redcare shows faster promises when the model is confident, it should also show longer promises when the model indicates higher delivery risk. Business is concerned that showing longer delivery times, even for a smaller share of products, could hurt conversion and make Redcare appear less competitive.

You have just joined as Senior Product Manager for Delivery & Shipping Experience. Your task is to decide how Redcare should introduce more precise delivery promises across the customer journey – including when the prediction is faster, unchanged, or slower than the current generic estimate.

Current: broad 1-3 working days	Proposed: precise promises on PDP, basket, and checkout
ML API: Sevenum warehouse only; faster for ~30% of eligible products	Last Mile signal: some cohorts need 2-3 or 3-5 working days
Main risk: early promise may change due to basket, address, cut-off, or carrier	Customer Service signal: late or unclear delivery already drives WISMO contacts

The table below shows a simplified backtest of the delivery prediction API. The API was run in the background on live order data, without changing the customer-facing delivery promise. For each order, the model estimated which delivery window it would have shown to the customer if the new promise experience had been live. These estimated promise windows were then compared with the actual delivery times.

Illustrative model output (anonymized, simplified and directional)

Window shown	Share	Within promise	Delivered later
1-2 days	30.3%	93.4%	6.6%
2-3 days	34.1%	94.5%	4.8%
1-3 days	24.9%	96.5%	3.5%
2-4+ days	7.0%	98.7%	0.0%
Total / weighted avg.	100.0%	95.3%	4.3%

Tasks

(a) Business says the conversion case is clear and wants to launch faster promises quickly. Last Mile argues that the model should also be used when it predicts a longer but more realistic delivery window. What is your position? Define what you would and would not accept – and on what conditions you would agree to a limited test versus a broader rollout. How do you hold the

line without damaging the relationship with Business, Data, Operations, Last Mile, and Customer Service?

(b) Design the experiment you would run to make this decision with confidence. What are the success and guardrail metrics, what sample and duration do you need, and what results would cause you to recommend for or against expansion? Cover both sides of the model: faster promises that may improve conversion, and longer promises that may reduce late deliveries and WISMO contacts but could hurt conversion.

(c) Now assume Redcare has agreed to test a more precise promise experience. Define how the promise should behave across the customer journey and how the customer experience should handle uncertainty. Where does Redcare move from a soft estimate to a confirmed order promise? How should PDP, basket, checkout, order confirmation, and tracking differ? How should mixed baskets, missing address, low confidence, missed cut-off, API failure, and operational disruption be handled?

2 Build It: Vibe-Code a Prototype

THE BRIEF

Delivery Promise Experience - a fictional Redcare feature

Today, a customer may see a broad delivery estimate such as "1-3 working days" throughout the journey. Redcare wants to explore a more precise and more helpful delivery promise experience without creating false certainty or operational risk.

We want to explore what a more precise delivery promise experience could look like: something that helps customers understand when their order is likely to arrive, handles uncertainty across products, baskets, warehouses and carriers, and builds trust — without overpromising or exposing operational complexity to the customer.

Your task: use AI-assisted coding tools to build a basic prototype of this fictional feature. It does not need to be production-ready, and you can choose any fidelity level. The recommended minimum flow is PDP, basket, and checkout. Order confirmation or tracking can be added if useful.

Suggested tools: Cursor · v0 by Vercel · Lovable · Claude · Figma + AI · Any tool you prefer

Tasks

(a) Submit a link to the working prototype alongside a brief writeup (1/2-1 page): the product decision you are most proud of, the biggest trade-off you made, and one thing you would do differently with more time.

(b) In the review session, be prepared to walk us through the prototype live and explain what you chose to show at which touchpoint and explain your choices.

(c) We are not evaluating code quality or high fidelity. We are evaluating product thinking: how you scoped a complex delivery experience under time pressure, what you chose to show and hide, and how fluently you use AI tools as part of your craft.

What We're Evaluating

Delivery Promise Judgement Do you understand how delivery promises depend on warehouse, carrier, cut-off time, country, stock, product type, basket composition, postcode, operational capacity, and last-mile performance?	Funnel & UX Judgement Can you distinguish between upper-funnel motivation and lower-funnel commitment? Can you handle the difference between product-level estimates and order-level promises?
Product System Thinking Can you translate operational complexity into clear customer-facing rules and scalable product logic, including fallback states, confidence thresholds, and mixed baskets?	Data & Experimentation Can you define success and guardrails beyond conversion uplift? Do you measure promise accuracy, contact rate, customer satisfaction, operational impact, and cost-to-serve?
Customer Trust & Health Context Do you recognise that a delivery promise is not just a conversion lever, but part of the trust experience in an e-pharmacy context?	AI Tool Fluency Do you use AI tools with confidence and intentionality — as an extension of your craft — or do the outputs feel unowned?

Submission Guidelines

- > Question 1: slide deck or structured written document. Depth over breadth.
- > Question 2: prototype link + short writeup (1/2-1 page).
- > State your assumptions explicitly - we intentionally left data gaps.
- > You do not need to answer every sub-question exhaustively. Prioritise where you have genuine conviction.
- > Review session: 60 min total - ~15 min Q1 + 15 min Q&A, ~15 min live prototype walk-through + 15 min Q&A.

Note on AI use: Use AI tools freely — for writing, analysis, and especially the prototype. What we cannot evaluate if AI does the thinking for you is domain depth, judgement under ambiguity, and the ability to hold a defensible position. Make it yours.